By construction, we get that

$$\mathbf{y}^1 = \mathrm{ReLU}\big(\mathbf{W}^1\mathbf{x}_t^2 + \mathbf{b}^1\big) \tag{118a}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathrm{ReLU}(\ell(t) + 1 + 1 - (t+1)) \\ \mathrm{ReLU}(2 - (t+1)) \end{pmatrix} \tag{118b}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathrm{ReLU}(\ell(t) + 1 - t) \\ \mathrm{ReLU}(1 - t) \end{pmatrix} \tag{118c}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathbb{1}\{\ell(t) \geq t\} \\ \mathbb{1}\{t < 1\} \end{pmatrix} \tag{118d}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathbb{1}\{\ell(t) = t\} \\ \mathbb{1}\{t = 0\} \end{pmatrix} \tag{118e}$$

$$\mathbf{y}^2 = \mathrm{ReLU}\big(\mathbf{W}^2\mathbf{y}^1 + \mathbf{b}^2\big) \tag{119a}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathrm{ReLU}(\mathbb{1}\{\ell(t) = t\} - \mathbb{1}\{t = 0\}) \\ \mathbb{1}\{t = 0\} \end{pmatrix} \tag{119b}$$

$$\tag{119c}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ \mathbb{1}\{\mathbb{1}\{\ell(t) = t\} \,\&\, \mathbb{1}\{t > 0\}\} \\ \mathbb{1}\{t = 0\} \end{pmatrix} \tag{119d}$$

$$\mathbf{y}^3 = \mathrm{ReLU}\big(\mathbf{W}^3\mathbf{y}^2 + \mathbf{b}^3\big) \tag{120a}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ \mathrm{ReLU}\big([\![v_{t-1}]\!] - (\mathbb{1}\{\ell(t) = t \wedge t > 0\} + \mathbb{1}\{t = 0\})\,\mathbf{1}_{|\Gamma|}\big) \\ [\![a_{t-1}]\!] \\ \mathrm{ReLU}\big([\![\sqcup]\!] + \mathbb{1}\{\ell(t) = t \wedge t > 0\}\,\mathbf{1}_{|\Gamma|} - \mathbf{1}_{|\Gamma|}\big) \\ \mathrm{ReLU}\big([\![\bot]\!] + \mathbb{1}\{t = 0\}\,\mathbf{1}_{|\Gamma|} - \mathbf{1}_{|\Gamma|}\big) \end{pmatrix} \tag{120b}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ \mathbb{1}\{\neg(\ell(t) = t \wedge t > 0) \wedge \neg(t = 0)\}\,[\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ \mathbb{1}\{\ell(t) = t \wedge t > 0\}\,[\![\sqcup]\!] \\ \mathbb{1}\{t = 0\}\,[\![\bot]\!] \end{pmatrix} \tag{120c}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ \mathbb{1}\{(\ell(t) < t \vee t = 0) \wedge t > 0\}\,[\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ \mathbb{1}\{\ell(t) = t \wedge t > 0\}\,[\![\sqcup]\!] \\ \mathbb{1}\{t = 0\}\,[\![\bot]\!] \end{pmatrix} \tag{120d}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ \mathbb{1}\{\ell(t) < t \wedge t > 0\}\,[\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ \mathbb{1}\{\ell(t) = t \wedge t > 0\}\,[\![\sqcup]\!] \\ \mathbb{1}\{t = 0\}\,[\![\bot]\!] \end{pmatrix} \tag{120e}$$

$$\mathbf{y}^4 = \mathrm{ReLU}\big(\mathbf{W}^4\mathbf{y}^3 + \mathbf{b}^4\big) \tag{121a}$$

$$= \begin{pmatrix} [\![q_t]\!] \\ \mathbb{1}\{\ell(t) < t \wedge t > 0\}\,[\![v_{t-1}]\!] + \mathbb{1}\{\ell(t) = t \wedge t > 0\}\,[\![\sqcup]\!] + \mathbb{1}\{t = 0\}\,[\![\bot]\!] \\ [\![a_{t-1}]\!] \end{pmatrix} \tag{121b}$$

$$\stackrel{\text{def}}{=} \begin{pmatrix} [\![q_t]\!] \\ [\![w_t]\!] \\ [\![a_{t-1}]\!] \end{pmatrix} \tag{121c}$$

Since the three logical expressions in Eq. (121b) are complementary, it holds that the second component of $\mathbf{y}^4$ holds either the value of v_{t-1}, $\sqcup$, or $\bot$ depending on the value of $\ell(t)$ and t. This is exactly the symbol that will be read by the PTM at time step t, i.e., s_t: If $\ell(t) < t$ (which means that cell $c(t)$ has been visited before), then $s_t = v_{t-1}$; if $\ell(t) = t$ and $t > 0$ (which means that cell $c(t)$ has not been visited before, and $t > 0$), then $s_t = \sqcup$; and if $t = 0$ (meaning that the PTM just started executing and is still reading the initial symbol $\bot$), then $s_t = \bot$. By the construction of $\mathbf{W}^5$ and $\mathbf{b}^5$, we get that $\mathbf{y}^5 = [\![q_t, s_t, a_{t-1}]\!]$. ■

Lemma F.9 (Correctness of the sampling step)**.** *Define the output matrix* $\mathbf{E} \in \mathbb{R}^{|Q||\Gamma||\overline{\Sigma}_\varepsilon||A||A| \times |Q||\Gamma||A|}$ *as*

$$E_{(q',v,\overline{y},a',a),(q,s,a)} \stackrel{\text{def}}{=} \begin{cases} p(q', v, \overline{y}, a', a \mid q, s) & \textbf{\textit{if}}\ \overline{y} \neq \text{EOS} \\ p(q_\varphi, v, \overline{y}, a', a \mid q, s) & \textbf{\textit{otherwise}} \end{cases} \tag{122}$$

for $q, q' \in Q$, $v \in \Gamma$, $\overline{y} \in \overline{\Sigma}_\varepsilon$, $a \in A$, $a' \in \{-1, 1\}$, *and* $s \in \Gamma$. *Then, it holds that*

$$\mathbf{E}\, F\left(\mathbf{x}_t^2\right) = p(\cdot \mid q_t, s_t)\,. \tag{123}$$

Proof. Follows directly from Lemma F.8 and the construction of $\mathbf{E}$:By Lemma F.8, we have that

$$F\left(\mathbf{x}_t^2\right) = [\![q_t, s_t, a_{t-1}]\!]. \tag{124}$$

By construction of $\mathbf{E}$, we have that

$$\left(\mathbf{E}\, F\left(\mathbf{x}_t^2\right)\right)_{(q',v,\overline{y},a',a)} = \left(\mathbf{E}\, [\![q_t, s_t, a_{t-1}]\!]\right)_{(q',v,\overline{y},a',a)} \tag{125a}$$

$$= \mathbf{E}_{(q_t,s_t,a_{t-1}),(q',v,\overline{y},a',a)} \tag{125b}$$

$$= \begin{cases} p\left(q', v, \overline{y}, a', a \mid q, s\right) & \textbf{if } \overline{y} \neq \text{EOS} \\ p\left(q_\varphi, v, \overline{y}, a', a \mid q, s\right) & \textbf{otherwise} \end{cases}. \tag{125c}$$

■